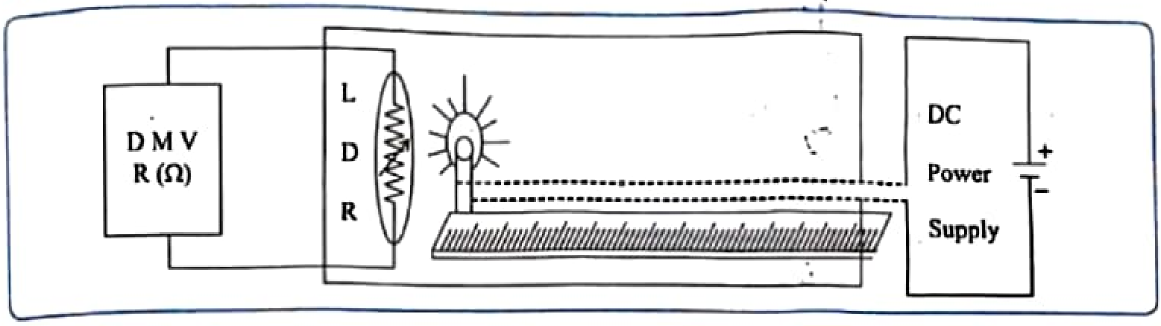
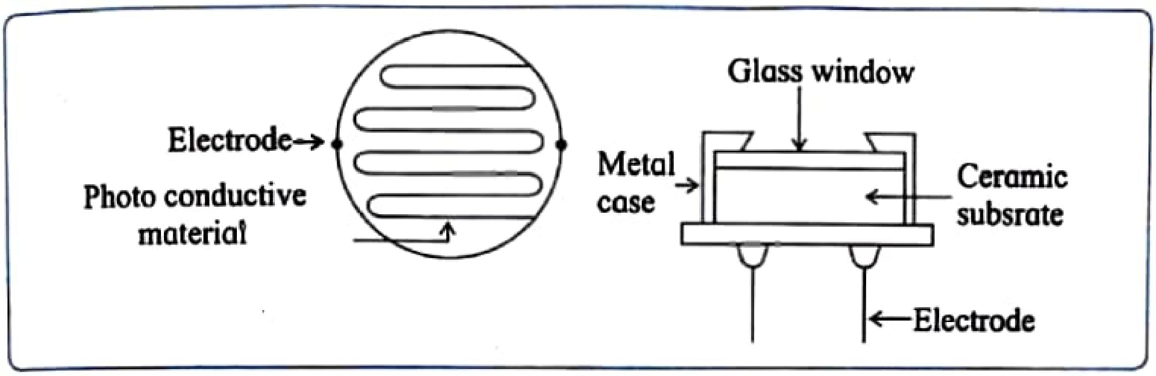
ACTIVITY NO. 9
LIGHT DEPENDENT RESISTOR

Aim : To study the effect of intensity of light on light dependent resistor (L.D.R.)

Apparatus : L.D.R., filament lamp (16V, 3W), meter scale, D.C. power supply, Digital multimeter. (DMM)

Circuit Diagram :





Theory :

The photo conductive cell is a two terminal semiconductor device whose resistance varies with the intensity of the incident light i.e. the resistance of the material decreases with exposure to light. Hence it is also called Light Dependent Resistor (LDR).

Light Dependent Resistors are fabricated from semiconductor materials, e.g. cadmium sulphide (CdS), cadmium selenide (Cd Se), cadmium - sulpho selenide (Cd_2S Se), lead sulphide (PbS).

When energy ($E = h\nu$) of the incident photon is greater than the energy band gap (Φ) between the valence band and the conduction band, electrons will be excited from the valence band to the conduction band. Electron - hole pairs thus generated will serve as charge carriers. Since conductivity of the material increases, its resistance decreases.

Procedure :

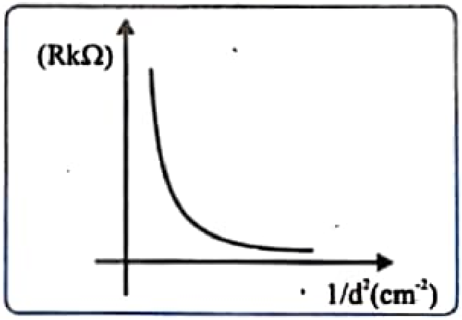
1. The given enclosed box consists of LDR fitted at its one end.
2. A filament bulb is fitted at one end of a movable half meter scale in line with LDR.
3. Connect the circuit as shown in the diagram
4. Keep the power supply knob on 10V, and switch ON the power supply.
5. Initially keep the lamp and the LDR very close to each other, measure the distance 'd' between the LDR and the lamp. Note down the corresponding resistance on DMM.

6. Increase distance 'd' by 3 cm each time and note down the corresponding resistance.
7. Plot a graph of R against $1/d^2$

Observations:

Obs. No	R (kΩ)	d (cm)	d^2 (cm^2)	$1/d^2$ (cm^{-2})
1	0·45	6	36	0·02778
2	0·71	10	100	0·01000
3	1·00	15	225	0·00400
4	1·35	20	400	0·00250
5	1·66	25	625	0·00160
6	2·06	30	900	0·00110
7	2·55	35	1225	0·00080
8	3·00	40	1600	0·00063
9	3·36	44	1936	0·00052
10	3·76	50	2500	0·00040

Graph: A graph of R aginst $1/d^2$



Conclusion:

1. The intensity of light is inversely proportional to the square of distance.
2. As the intensity of light decreases, the resistance of LDR increases. The graph of R against $1/d^2$ shows the exponential decrease.

Precautions :

1. The light from outside should not fall on the LDR.
2. The box is blackened from inside, so that light scattered or reflected from interior of enclosure does not fall on LDR.

Questions

1) What is the relationship between intensity of light and resistance?
Ans.
The resistance of an LDR decreases as the intensity of light increases. Thus, resistance is inversely proportional to the intensity of light.

2) State the applications of LDR.
Ans.
LDR is used in smoke detectors, optical coding systems, security alarm systems, and proximity switches. It is also used in automatic street lighting and other light-sensitive control circuits.

Remark and sign of teacher : ....

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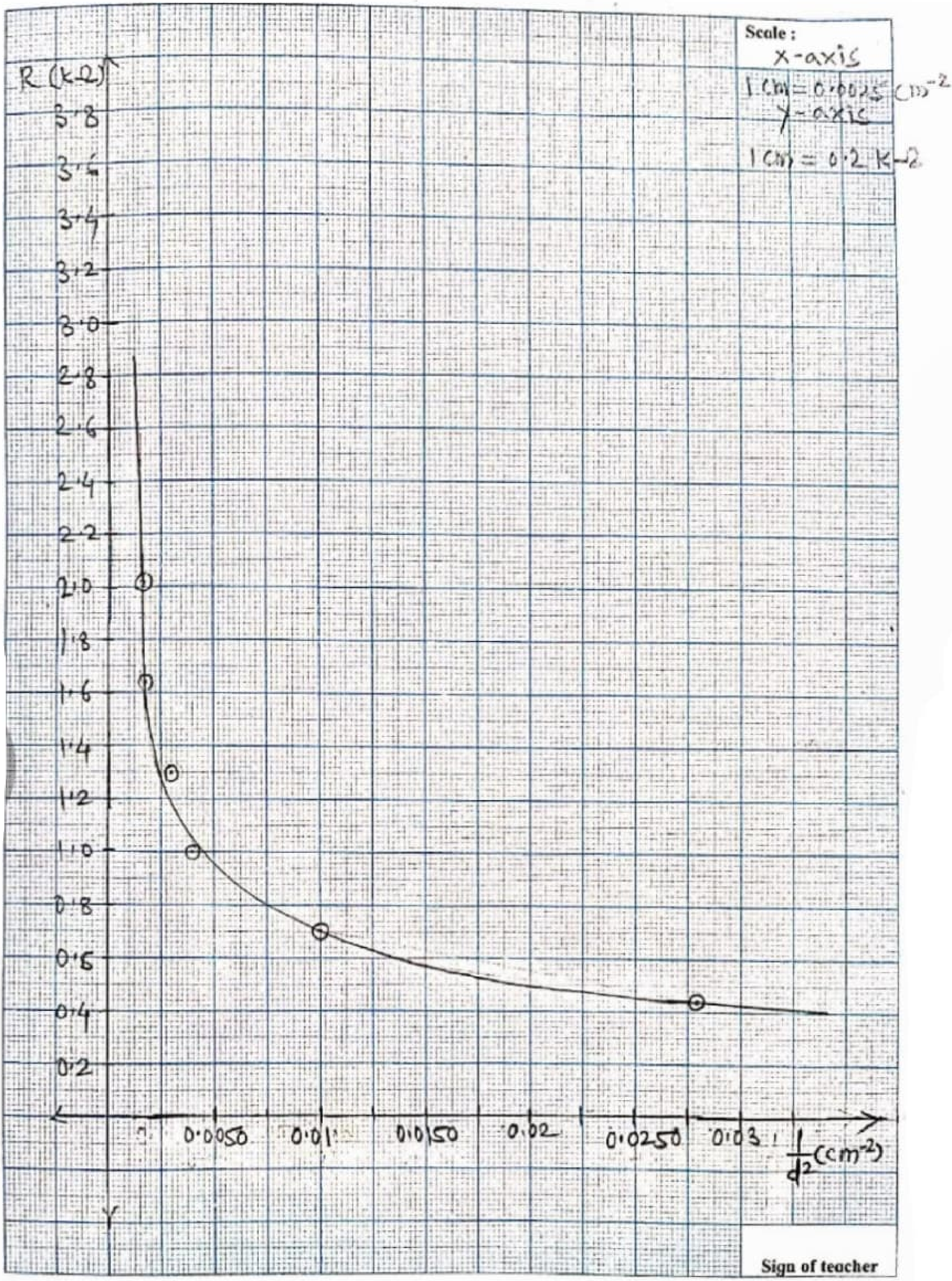

Scale :
x-axis
1 cm = 0.0025 cm^{-2}
y-axis
1 cm = 0.2 kΩ
R (kΩ)
3.8
3.6
3.4
3.2
3.0
2.8
2.6
2.4
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0050
0.01
0.0150
0.02
0.0250
0.03
$\frac{1}{d^2}$ (cm^{-2})
Y
Sign of teacher